

NB04: Context-free Languages

Language Processing

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✓ 4.1 Context-free Languages

- *Context-free languages* are the second class in the Chomsky hierarchy
 - They strictly extend the expressive power of regular languages
- Regular languages are well suited for local patterns
 - fixed repetition
 - simple alternation
 - bounded memory
- But recognizing some languages require unbounded memory, for example:
 - "the same number of symbols" (e.g. $a^n b^n$)
 - nested or recursive structures (parentheses, blocks, expressions)
- These languages cannot be described by regular expressions (REs) nor finite automata (FAs)
- Instead, context-free languages are the languages recognized by **pushdown automata** (PDAs)
 - PDAs extend FAs with a symbol **stack**
- They are typically described formally by **context-free grammars** (CFGs)
- These languages play a central role in *syntactic analysis*
 - the syntax of most programming languages is context-free
- There exist languages that are not regular nor context-free
 - recognizing them requires more powerful models (e.g. Turing machines)
 - we will not go beyond context-free languages in these notes

Context-free language example: A language of nested parenthesis

- Consider a language that considers only $($ and $)$ symbols
- Every opening $($ must be matched by a closing $)$
 - $((()())())$ ✓
 - $((()))$ ✗
 - $((()())$ ✗
- Can you think of an RE to recognize this language?
- Any proper computer language requires syntactic rules of this kind
- Rule description:
 - Any parenthesis that opens, must be eventually closed
 - We must somehow "memorize" how many parenthesis remain to be closed at each point
- How can we **formally specify** these rules?
- How can we **efficiently recognize** such valid inputs?

4.2 Context-free Grammars

- Context-free grammars (CFGs) provide a rule-based description of context-free languages
- They are mainly used in the specification of the the syntax of programming languages, enabling parsing and syntax tree construction

Formal grammars

- A (formal) *grammar* consists of:
 - symbols we want to generate (terminals)
 - auxiliary symbols used during generation (nonterminals)
 - rules that rewrite nonterminals (productions)
- They are formal in contrast with natural language grammars

Grammar example: Bitstrings with at least one 1

```

S → A1A
A → A1
A → A0
A → ε

```

Or, grouping together production for the same nonterminal:

```

S → A1A
A → A1 | A0 | ε

```

Definition: Grammar

- A *grammar* G is a tuple (Σ, N, P, S_0) where:
 - Σ is the set of *terminal* symbols (the alphabet)
 - N is a set of *nonterminal* symbols (the grammar's variables)
 - P is a set of *productions* that state how nonterminal symbols can be expanded into sequences of terminal and nonterminal symbols
 - $S_0 \in N$ is the *start* symbol

Derivations and parse trees

- A **derivation** is a sequence of applications of productions that expand nonterminals
 - One derivation step is denoted by $\Rightarrow$
 - Zero or more derivation steps are denoted by $\Rightarrow^*$
- We say that a derivation is *leftmost* / *rightmost* each step always expands the *leftmost* / *rightmost* nonterminal
- A word $w \in \Sigma^*$ (a sequence of terminal symbols) is **generated** by a grammar G with start symbol S if $S \Rightarrow^* w$
- A **parse tree** is a hierarchical representation of a derivation:
 - the root is the start symbol
 - internal nodes are nonterminals
 - leaves are terminals (or ϵ)
 - reading the leaves from left to right yields the derived word
- The set of words generated by a grammar defines a language



Derivation example: Bitstrings with at least one 1

- The grammar shown above has terminals $\Sigma = \{0, 1\}$ and nonterminals $N = \{S, A\}$, with start symbol $S_0 = S$
- The productions P are those already shown

```
S → A1A
A → A1 | A0 | ε
```

Is 0110 derivable by this grammar?

Leftmost derivation:

```
S ⇒ A1A ⇒ A11A ⇒ A011A ⇒ ε011A ⇒ 011A0 ⇒ 011ε0 = 0110
```

Since $S \xRightarrow{*} 0110$ and S is the start symbol, 0110 belongs to the language generated by the grammar.

Regular grammars

- The regular languages that we have been expressing with REs can also be specified with formal grammars
- In particular, all regular languages can be generated by a *regular grammar*
- A regular grammar must follow these rules:
 - Only one nonterminal symbol on the left-hand side of productions
 - At most one nonterminal symbol in the right-hand side of productions, and always as the last symbol (or always as the first symbol)



Regular grammar example: Valid identifiers

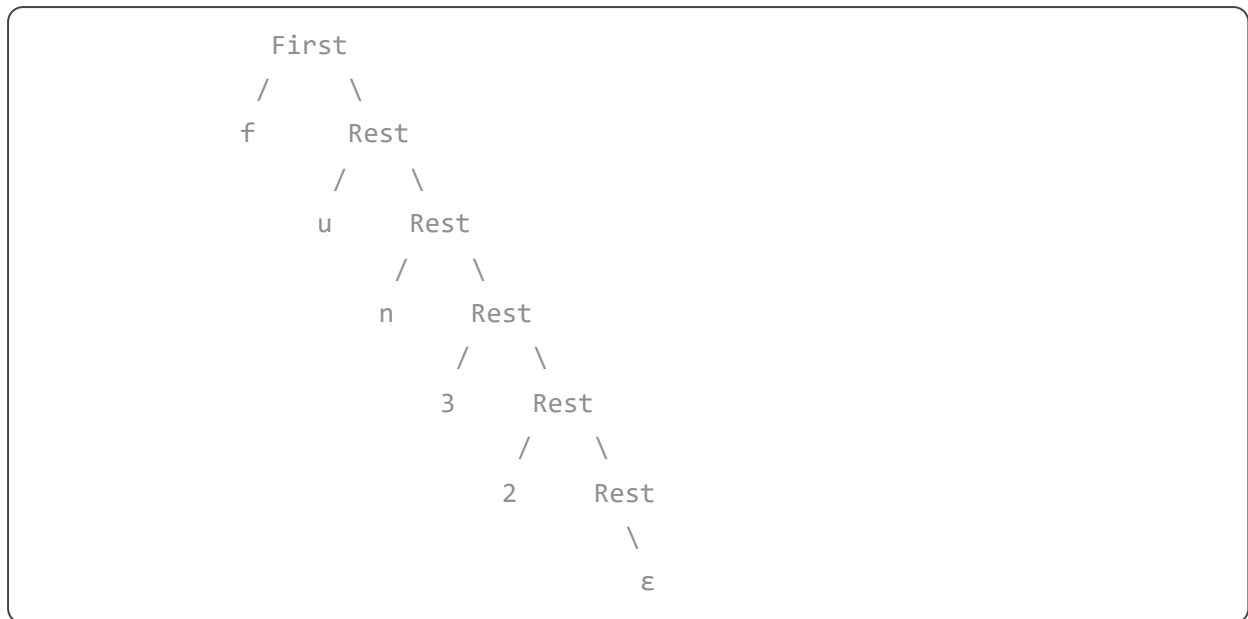
Grammar with $\Sigma = \{a, b, \dots, Z, O, \dots, 9, _ \}$, $N = \{First, Rest\}$, $S_0 = First$, and the following productions:

```
First → aRest | bRest | ... | ZRest | _Rest
Rest  → aRest | bRest | ... | ZRest | _Rest | 0Rest | ... | 9Rest | ε
```

Word $fun32$ can be derived as:

```
First ⇒ fRest ⇒ fuRest ⇒ funRest ⇒ fun3Rest ⇒ fun32Rest ⇒ fun32ε = fun32
```

Or constructing a parse tree:



Context-free grammars

- *Context-free grammars* (CFGs) allow the specification of context-free languages
- They must follow this rule:
 - Every production only has one nonterminal symbol in the left-hand side
- This is why they are called context-free
 - A nonterminal can always be expanded by the same productions, regardless of the surrounding symbols



CFG example: A language of nested parenthesis

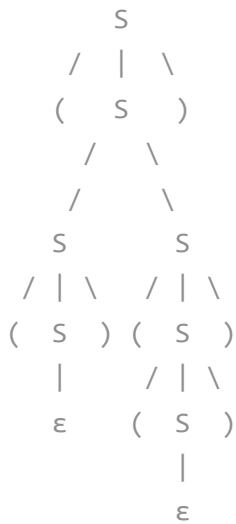
Grammar with $\Sigma = \{ (,) \}$, $N = \{ S \}$, $S_0 = S$, and the following productions:

$$S \rightarrow (S) \mid SS \mid \epsilon$$

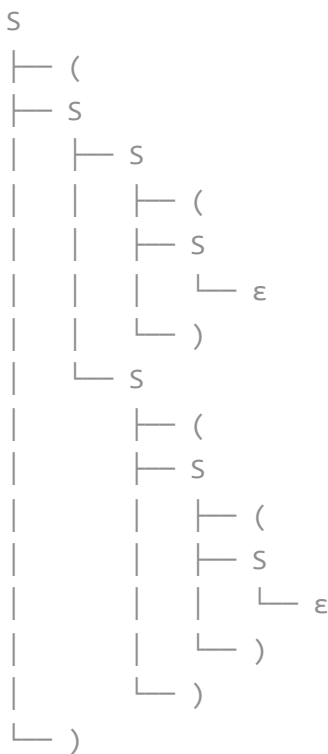
Word $((()((())))$ can be (leftmost) derived as:

$$S \Rightarrow (S) \Rightarrow (SS) \Rightarrow ((S)S) \Rightarrow (()S) \Rightarrow (() (S)) \Rightarrow (() ((S))) = (()((())))$$

Or constructing a parse tree:



Or in a tree style that is to draw:



- CFG example: A language for arithmetic S-exprs

In [Lisp](#)-style languages, S-exprs (symbolic expressions) represent computations using prefix notation and parentheses. In this notation:

- The operator appears first, followed by its operands
- Operands can be numbers or nested expressions

Example S-exprs:

- $(+ \ 1 \ 2)$
- $(\ast \ 3 \ 4)$

- $(+ (* 2 3) 5)$
- $(- (+ 4 5) (* 2 3))$

$\Sigma = \{ (,), +, -, *, /, 0, 1, \dots, 9 \}$, $N = \{ \text{Expr}, \text{Op}, \text{Num}, \text{Dig} \}$, $S_0 = \text{Expr}$

```
Expr → Num | ( Op Expr Expr )
Op   → + | - | * | /
Num  → Dig | Dig Num
Dig  → 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
```

Leftmost derivation of $((* (+ 2 3) 5))$:

```
Expr ⇒ ( Op Expr Expr ) ⇒ ( * Expr Expr ) ⇒ ( * ( Op Expr Expr ) Expr )
      ⇒ ( * ( + Expr Expr ) Expr ) ⇒ ( * ( + Num Expr ) Expr ) ⇒ ( * ( + Dig Expr )
      ⇒ ( * ( + 2 Expr ) Expr ) ⇒ ( * ( + 2 Num ) Expr ) ⇒ ( * ( + 2 Dig ) Expr )
      ⇒ ( * ( + 2 3 ) Expr ) ⇒ ( * ( + 2 3 ) Num ) ⇒ ( * ( + 2 3 ) Dig )
      ⇒ ( * ( + 2 3 ) 5 )
```

Parse tree for $((* (+ 2 3) 5))$:

```
Expr
├── (
│   ├── Op
│   │   └── *
│   ├── Expr
│   │   ├── (
│   │   │   ├── Op
│   │   │   │   └── +
│   │   │   ├── Expr
│   │   │   │   ├── Num
│   │   │   │   │   └── Digit
│   │   │   │   │       └── 2
│   │   │   └── Expr
│   │   │       ├── Num
│   │   │       │   └── Digit
│   │   │       │       └── 3
│   │   └── )
│   ├── Expr
│   │   ├── Num
│   │   │   └── Digit
│   │   │       └── 5
│   └── )
```

✂ CFG exercise: Language for arithmetic expressions

- Write a CFG to recognize standard arithmetic expressions over integer numbers

- $(10 + 20) * 30$ ✓
- $10 + -20$ ✓
- $(10 + 20 * 30)$ ✗
- $10 \ 20 * 30$ ✗
- $10 + * 20$ ✗

$\Sigma = \{ +, *, /, -, (,), 0, 1, \dots, 9 \}$
 $N = \{ \text{Expr, Op, Num, Dig} \}$
 $S_0 = \text{Expr}$

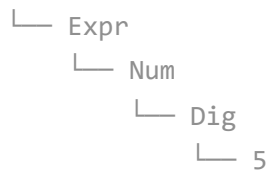
```
Expr → Expr Op Expr | -Expr | (Expr) | Num
Op   → + | - | * | /
Num  → Dig | Dig Num
Dig  → 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
```

Leftmost derivation of $(2 + 3) * 5$:

```
Expr ⇒ Expr Op Expr ⇒ (Expr) Op Expr ⇒ (Expr Op Expr) Op Expr ⇒ (Num Op Expr)
      ⇒ (Dig Op Expr) Op Expr ⇒ (2 Op Expr) Op Expr ⇒ (2 + Expr) Op Expr ⇒ (2
      ⇒ (2 + Dig) Op Expr ⇒ (2 + 3) Op Expr ⇒ (2 + 3) * Expr ⇒ (2 + 3) * Num ⇒
      ⇒ (2 + 3) * 5
```

Parse tree for $(2 + 3) * 5$:

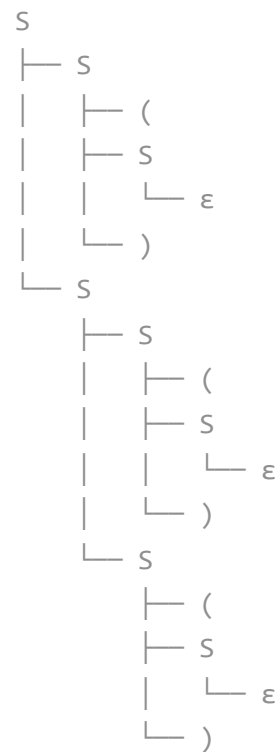
```
Expr
├── Expr
│   ├── (
│   │   ├── Expr
│   │   │   ├── Expr
│   │   │   │   ├── Num
│   │   │   │   │   ├── Dig
│   │   │   │   │   │   2
│   │   │   │   ├── Op
│   │   │   │   │   ├── +
│   │   │   │   ├── Expr
│   │   │   │   │   ├── Num
│   │   │   │   │   │   ├── Dig
│   │   │   │   │   │   3
│   │   │   │   │   └── )
│   │   └── Op
│   │   │   └── *
│   └── Expr
```

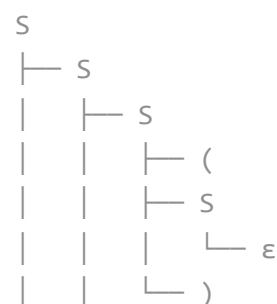
✓ Ambiguous grammars

- Let us get back to the language for nested parenthesis
- For many words, there are alternative derivations that give rise to different parse trees
- Two leftmost derivations and parse trees for $((()()))$

◦ $S \Rightarrow SS \Rightarrow (S)S \Rightarrow ()S \Rightarrow ()SS \Rightarrow ()(S)S \Rightarrow ()()S \Rightarrow ()()(S) \Rightarrow ()()()()$



◦ $S \Rightarrow SS \Rightarrow SSS \Rightarrow (S)SS \Rightarrow ()SS \Rightarrow ()(S)S \Rightarrow ()()S \Rightarrow ()()(S) \Rightarrow ()()()()$





- When, for any word, there are two different leftmost (or rightmost) derivations, or two different parse trees, we say that the grammar is **ambiguous**
- Ambiguity does not affect the language generated by a grammar (the same set of words is derivable)
- But ambiguous grammars are unsuitable for syntactic analysis
 - The same program would have multiple valid meanings
- Often, ambiguous grammars can be rewritten into an equivalent unambiguous version
 - A version that generates the same set of words, but only with one parse tree for each word
- However, there are languages that are *inherently ambiguous*
 - There is no unambiguous grammar that generates such a language
 - E.g., $\{a^i b^j c^k \mid i = j \vee j = k, i, j, k > 0\}$



Ambiguity example: A language for nested parenthesis

- The following is an equivalent, but unambiguous, version of the nested parenthesis CFG:

$$S \rightarrow S(S) \mid \epsilon$$

- Leftmost derivation of $((())())$:
 - $S \Rightarrow S(S) \Rightarrow S(S)(S) \Rightarrow S(S)(S)(S) \Rightarrow (S)(S)(S) \Rightarrow (())(S)(S) \Rightarrow (())(S)() \Rightarrow (())()()$
- Parse tree for $((())())$:

```

S
├ S
│   ├── S
│   │   ├── S
│   │   │   └ ε
│   │   ├── (
│   │   │   ├── S
│   │   │   │   └ ε
│   │   │   └ )
│   │   ├── (
│   │   │   ├── S
│   │   │   │   └ ε
│   │   │   └ )
│   ├── (
│   │   ├── S
│   │   │   └ ε
│   │   └ )
├ (
├ S
├   └ ε
└ )

```

✂ Ambiguity exercise: Language for arithmetic expressions

- The CFG presented above for arithmetic expressions is highly ambiguous
- Identify why, and rewrite the grammar to a unambiguous version

- $1 + 2 * 3$

```

          Expr
        /  |  \
      Expr Op Expr
    /  |  \  |
  Expr Op Expr * Num
    |  |  |  |
  Num + Num   Dig
    |  |  |
  Dig Dig    3
    |  |
    1  2

```

```

          Expr
        /  |  \
      Expr Op Expr
    /  |  \  /  |  \
  Num + Expr Op Expr
    |  |  |  |  |

```

Dig	Num	*	Num
1	Dig		Dig
	2		3

- The issue here is the lack of *operator precedence* rules:
 - The first tree interprets as $$(1 + 2) * 3 = 9$$, the second $$(1 + (2 * 3) = 7$$
- We can introduce new levels in the grammar to ensure that higher-precedence operators appear deeper in the parse tree

```

Expr  → Expr AddOp Expr | Term
Term  → Term MulOp Term | Factor
AddOp → + | -
MulOp → * | /
Factor → -Factor | (Expr) | Num
Num    → Dig | Dig Num
Dig    → 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

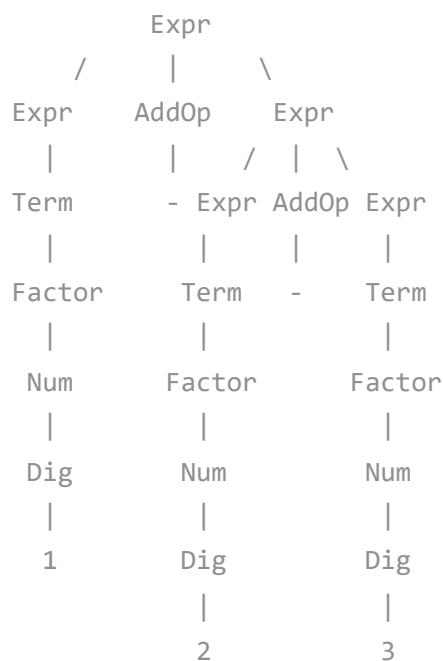
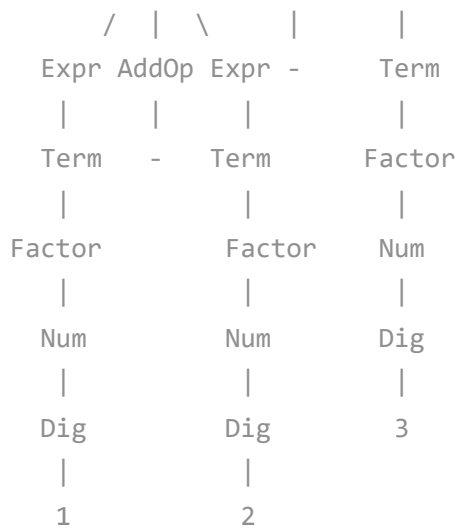
```

- There is now a single interpretation for $1 + 2 * 3$

Expr				
/		\		
Expr	AddOp	Expr		
Term	+	Term		
		/		\
Factor		Term	MulOp	Term
Num		Factor	*	Factor
Dig		Num		Num
1		Dig		Dig
		2		3

- $1 - 2 - 3$

Expr		
/		\
Expr	AddOp	Expr



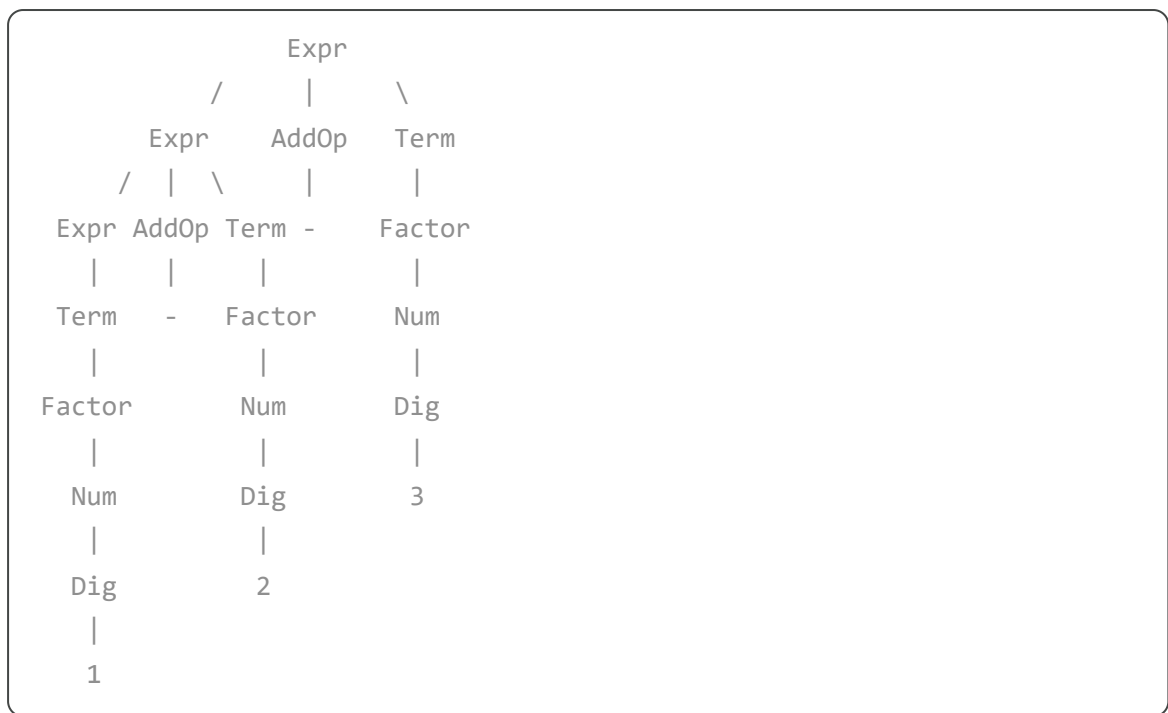
- The issue here is the lack of *associativity rules*:
 - The first tree interprets as $(1 - 2) - 3 = -4$, the second $1 - (2 - 3) = 0$
- We can control how recursion can be applied to force associativity rules

```

Expr  → Expr AddOp Term | Term
Term  → Term MulOp Factor | Factor
Factor → -Factor | (Expr) | Num
AddOp  → + | -
MulOp  → * | /
Num    → Dig Num | Dig
Dig    → 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

```

- There is now a single interpretation for $1 - 2 - 3$



✓ 4.3 CFG Recognition

Automata-view of context-free languages

- Much like regular languages are recognized by FAs, context-free languages are recognized by *pushdown automata* (PDAs)
- PDAs extend FAs with an auxiliary *stack*
- This stack provides auxiliary memory, but can only be accessed in a last-in first-out manner

✓ Definition: PDA

- A (non-deterministic) pushdown automaton (PDA) is a tuple $(Q, \Sigma, \Gamma, \delta, q_i, Z_0, F)$
 - Q is a finite set of states
 - Σ is the input alphabet
 - Γ is the stack alphabet
 - $\delta : Q \times \Sigma \cup \{\epsilon\} \times \Gamma \rightarrow \mathcal{P}(Q \times \Gamma^*)$ is a transition function
 - $q_i \in Q$ is the initial state
 - $Z_0 \in \Gamma$ is the initial stack symbol
 - $F \subseteq Q$ is a set of final (or accepting) states

- The transitions of a PDA read the top symbol of the stack, and add back a set of symbols
- A word $w \in \Sigma^*$ is processed by a PDA by successively applying δ to all the current states and symbols of w , starting in states $\{q_i\}$ and with a stack Z_0
- If one of the states reached after processing all symbols of w is final, w belongs to the language; otherwise, it does not
 - We call this this *acceptance by final state*
- Alternatively we can accept by *empty state*
 - In this case, final states are not considered, and a word is accepted if the stack is empty after processing the complete input

✚ Throughout the notebook, the `automata` Python module is used to draw and manipulate automata. This is not part of the syllabus.

✓ PDA example: A language of nested parenthesis

- Use the stack to memorize every open parenthesis
- When a closing parenthesis is seen, pop from the stack (if possible)
- If the bottom of the stack is seen, try to accept

```
# libraries to allow the visualization and manipulation of automata
!apt-get update -qq
!apt-get install -y -qq graphviz graphviz-dev
!pip install -q pygraphviz coloraide
!pip install -q automata-lib
```

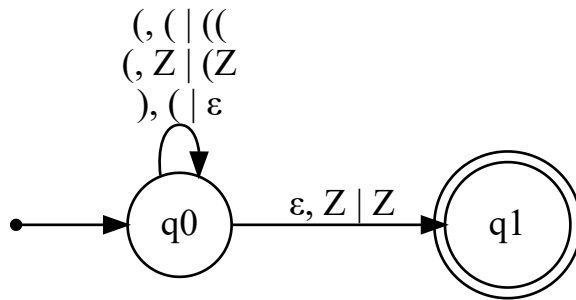
[Mostrar saída oculta](#)

- The following (non-deterministic) PDA accepts by final state

```
from automata.pda.npda import NPDA

NPDA(
    states={"q0", "q1"},
    input_symbols={"(", ")"},
    stack_symbols={"Z", "("},
    transitions={
        "q0": {
            "(": { "Z": {("q0", "(Z")}, "(": {("q0", "(")} },
            ")": { "(": {("q0", "")} },
            "Z": { "Z": {("q1", "Z")} } } },
    initial_state="q0",
    initial_stack_symbol="Z",
```

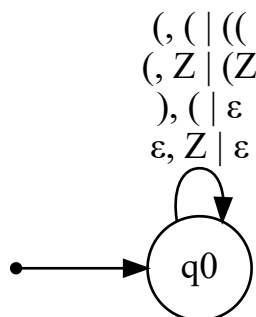
```
final_states={"q1"},
acceptance_mode="final_state").show_diagram(path = None)
```



- Alternatively, the following (non-deterministic) PDA accepts by empty stack

```
from automata.pda.npda import NPDA
```

```
NPDA(
    states={"q0"},
    input_symbols={"(", ")"},
    stack_symbols={"Z", "("},
    transitions={
        "q0": {
            "(": { "Z": {("q0", "(Z")}, "(": {("q0", "(")} },
            ")": { "(": {("q0", "")} },
            "": { "Z": {("q0", "")} } } },
    initial_state="q0",
    initial_stack_symbol="Z",
    final_states=set(),
    acceptance_mode="empty_stack").show_diagram(path = None)
```



CFG and PDA equivalence

- The CFGs and (non-deterministic) PDAs are equivalent
 - A CFG can be converted into a PDA automatically, and vice-versa
- Context-free languages are exactly those that can be described by a CFG or a non-deterministic PDA
- Note that, unlike FAs, non-deterministic PDAs and deterministic PDAs are not equivalent formalisms

- Deterministic PDAs can only recognize a subset of context-free languages

✓ Performance of membership tests

- Naively testing the membership for a word w in a context-free language may be exponential in the worst case
 - For instance, directly executing a non-deterministic PDA
- However, there are algorithms that can test membership in $O(n^3)$ time in the worst case
 - Cocke–Younger–Kasami (CYK) algorithm
 - Earley algorithm
- More efficient, linear algorithms exist for **subsets** of CFGs
 - We will focus a subset of context-free languages that can be recognized by *deterministic* PDAs
 - Programming languages in general fall in this subset

-- Nuno Macedo, 2026-03-10